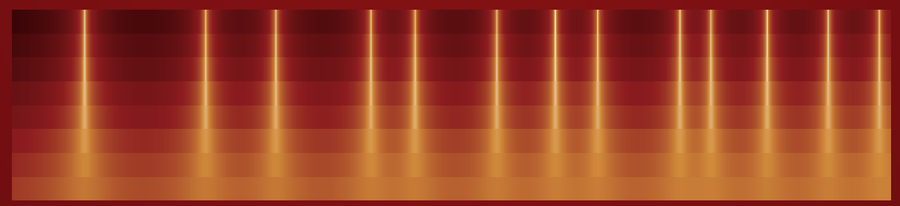
Where computation meets the Riemann Hypothesis

The Riemann Hypothesis can be stated as a positivity condition. The reciprocal Xi function is the Laplace transform of a generalised gamma convolution exactly when every zero lies on the critical line, and then its Thorin measure is the discrete measure on the squared ordinates. This book asks what computation can say about that condition.

It develops the Pólya route, which evaluates the Xi function in certified ball arithmetic from Pólya's kernel alone, and uses it to compute the Thorin measure at every resolution, to measure its distance to the measure demanded by RH, to read the zero density from a Thorin form of Turing's method, and to calibrate exactly what an off line zero would leave behind. The sine squared lemma emerges as the bridge from the primes to the zeros, and as the reason the primes alone cannot compute near the line.

Hitting times and Brownian boundaries, Keiper Li coefficients, the de Bruijn Newman flow, the Dedekind eta kernel of $L(s, \chi_{12})$ and a study of the reciprocal zeta function complete the picture, and Chapter 12 sets out a path to RH with every step marked proved, computed or open.

The last chapter reads RH as a statement of statistical mechanics: the Pólya law, whose characteristic function is the normalised Xi function, is ferromagnetic in the sense of Lee and Yang. With no hypothesis on the zeros, it builds a law within total variation 0.000002 of the Pólya law and proves by computer, in ball arithmetic, that this law is ferromagnetic. The code and data behind the book's computations are at github.com/VadimSokolov/computing_rh.



the same measure at eight resolutions, from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top)

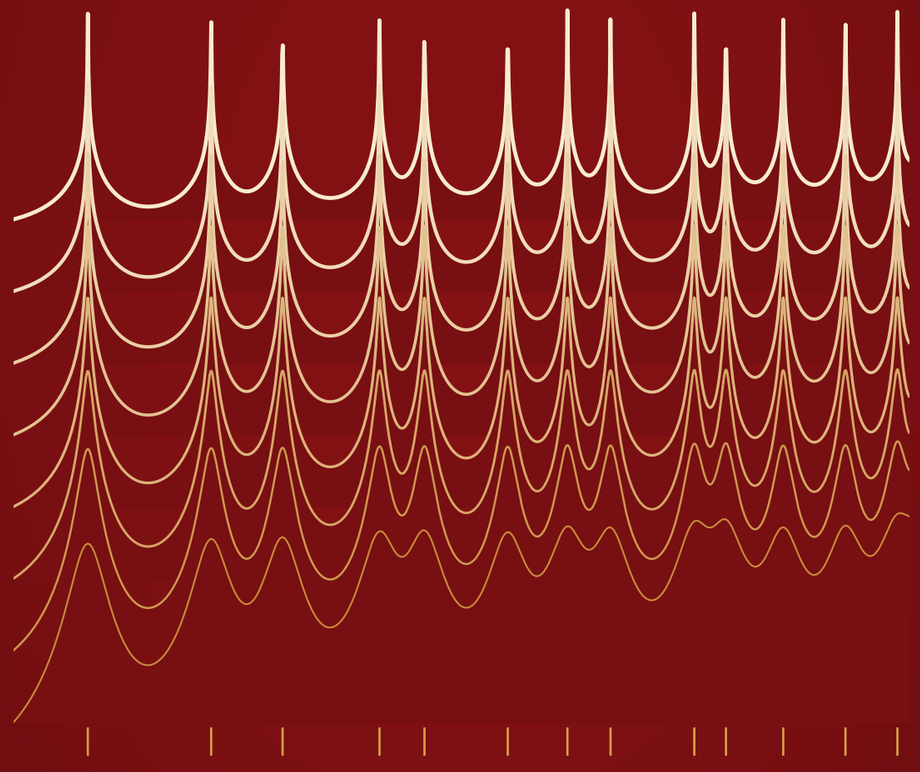
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Computing the Riemann Hypothesis

Thorin Measures, Pólya Densities and Hitting Times



the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ sharpening onto the zeros

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